Presentation:- Date

Effective
Interact Audience
Transmit with Clarity
Engage Audience
Understand listeners Mind.

Provide References.

Plan, Prepare, Practice, Present

Ask Question Who, Why When, Where,

Visuals
Voice
Appearance
Questions

Simple font
6 bullets max

Order
1) Objective
2) Background
3) Main
4) Summary
5) ~~Conclusion~~ Questions.

Computers:-
Calculating Machine
Arithmetic Ops
High Speed.

User
↳ Person interacting

↳ Input
Process. (4)
Output
Store

Data → Numbers Symbols etc
Raw facts
Preprocessing

Vs

Information
Processed data.
More Meaningful

Software:- Collection of computer instructions. Tells computer how to work.

Applications
Beyond Basic Ops
Entertainment.

System:-
Directly Operates System Hardware.
Platform for Apps.

Information Technology
Using systems to store, Retrieve, send info

Computer Evolution:-
Vacuum Tubes (1st Gen)
Transistors (2nd Gen)
Integrated circuits (3rd Gen)
VLSI, C, C++ (4th Gen)

5th gen → AI
6th → future.

LUCKY WRITE SERIES

Digital Computer:-

Date

Super — Mainframe — Mini — Micro — others

Mini → Server, Workstation.

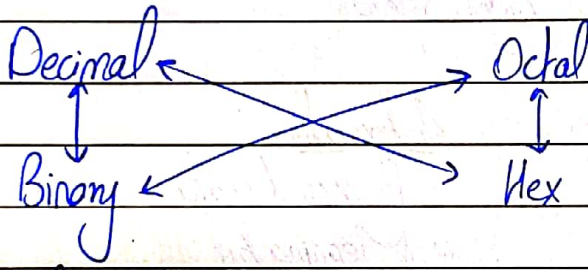
Micro → Laptop, phones.

others → Smartphones, Wearables, TVs → Android, Game Consoles

Organising Information:-

Remove Duplicates. (No Sync) (Waste Space)
Group in Sub-folders.
File naming Conventions. (Can't have special chars in file names)
Backup files.
Organisational plan.

Number Systems:-



Binary to Decimal $2^6\ 2^5\ 2^4\ 2^3\ 2^2\ 2^1\ 2^0$
1101001 1 (Multiply by weight)

Octal to Decimal $8^2\ 8^1\ 8^0$
724 (Multiply)

Hex to Decimal $16^2\ 16^1\ 16^0$
ABC

Binary from Decimal:- Keep dividing by 2. Start from bottom.

2	125	
2	62	1
2	31	0
2	15	1
2	7	1
2	3	1
2	1	1
	0	

$1111101_{(2)}$.

LUCKY WRITE SERIES

Date

Octal to Binary:- Convert each octal to 3 bit binary.

7 0 5 (8)
↙ ↓ ↘
111 000 101

$111000101_{(2)}$

Hex to Binary:-

4 bit each digit.

1 0 A F
↙ ↓ ↘ ↘
Binary. 0001 0000 1010 1111

$000100001010111_{(2)}$

Decimal to Octal:-

keep dividing by 8

8	1234	
8	154	2
8	19	2
8	2	3
	0	2

$2322_{(8)}$

Decimal to Hex:-

16	1234	
16	77	2
16	4	13 = D
	0	4

2 13 402

Binary to Octal:-

1011010111

1 011 010 111

Start from right. Make groups of 3

Binary to Hex

groups of 4

Binary Add:-

1
1+1 = 0
0+1 = 1
1+0 = 1

LUCKY WRITE SERIES

Fractions:- Date

$2^{1}\ 2^{-2}\ 2^{-3}\ 2^{-4}$

10.1011

$1 \times 2^{-4} = 0.0625$

$1 \times 2^{-3} = 0.125$

$0 \times 2^{-2} = 0$

$01 \times 2^{-1} = 0.5$

2.6875

Bits Binary digit

Bytes:- 8 bits grouped.

Nibble → 4 bits.

Word → 2 bytes.

Two's Complent | Negative number. (Sign Representation)

leftmost is only sign.

01100 (+12)

11100 (-12)

Emails:-

Online Communication:-

1) Cost saved.
2) Easy to use.
3) Accessible.
4) Globalisation.

Disadv

1) Need good internet.
2) Connection issues.
3) Misinterpretation.

Ethiquete

Correct way to behave.

Email Ethiquete

1) Always Subject. Recepient's name in greeting.
2) Don't write All Caps.
3) Proper Greetings. (Respect)
4) Not share C I.
5) Abbreviation + emojis careful.
6) Check email for proper grammar.

LUCKY WRITE SERIES

Date

Be mindful of email addres (Appropriate)

CC
↳ Carbon Courtesy

BCC
↳ Blind

Signature
Info about sender.

CBIS
contains
Hardware
Software
Databases

System:-
Components that interact to accomplish goals.

Process:-
Tasks performed to achieve outcome.

Information System:- Interrelated components. (collect, manipulate, store info)
Feedback mechanism.

CBIS:- Computer based.
Decisions to make business successful

BIS (Business)
Inter related procedures
IT infrastructure.
Generate & disseminate info

E-commerce B2B, C2C, B2C.
Business transaction electronically

M-commerce
↳ Wireless communication
Place order do business.

Internet. World's largest computer Network.

Intranet Internal network.
Within org

Extranet:- Access to intranet allowed to specific outsiders.

LUCKY WRITE SERIES

TPS :- Date

(Transaction Processing System)

Transaction → Business related exchange.

people + procedures + software + databases used to perform business transactions.

ERP :- (Enterprise Resource Planning)

Integrated programs for managing entire business.

Management Info System:-

Routine info to managers.
Provide Standard reports generated with info from TRP or ERP.

Input of data:-

Data entry
Editing
User Interface.

Structured — Strict format

Semi-Structured — some format, Some attributes.

Unstructured data — Simple text doc.

Databases:- help with large amount of info.
Tools for storing + searching + managing info.

Value of Info:-

Help achieve organisation's goals.

Date

Database:- Set of info in Computer.
large structure of persistent data (updated) (Queried)
Ease Speed of search.

- Store
- Update
- Retrieve.
- Organise
- Protect

DBMS:- (Management S) → MS Access

Software that controls info.

Provides
- Data Definition ← DDL
- Manipulation → DML
- Control ← DCL

Provides
- Data Independency.
- Security
- Integrity
- Concurrency.

Data Dictionary → (Tables, users, rules, views, index)
Info about database itself
Data about data (Metadata).
Who uses which data.

DCL	DML	DDL
Grant	INSERT delete	Create Alter.
Revoke.	Update	Drop

Files Based System:-

Files
Each file specific format

- No format
- Duplication.
- dependent.
- No security / recovery

Relational

- Rows → Tuples.
- Columns → Attributes.
- Tables → Relations.
- Data type → Domain.

Networks:- Date

Exchange of info b/w people ↔ computer.
Share printer.

Components:-
- Server
- Hub
- Switch
- Router
- LAN Cable
- Clients.
- NIC.

Connected Computers. | Cabling.
↳ Nodes

LAN:-
small network close by.

Every computer sees transmission but doesn't know if message is meant for it. So we use MAC Address.

Bandwith:- Rate transfer

Collision:-
if collision detected (wait random time) send again

Exponential backoff:-
Increase wait time after collision.

Domain name System:-
Translates Domain names to IP address

WWW:-
Each webpage has URL. (unique)
↳ fundamental block Web page

OS:-

Date

Information System:-
Process logically related tasks.
Defined outcome.

Knowledge:-
Understand set of info
Using info for a specific task.

Components of Info Sys:-
People → IT Professionals
Computer → Hardware + Software
Data → Raw
Network → communication.

Value of Info:-
Helps Decision Makers Achieve goals (Org).

CBIS:- [Hardware* Software* Databases*]
↓
IT and business Processes

Business Information System:-
Used to
1) Forecasting
2) Planning
3) Control
4) Co-ordination
5) Decision Making
6) Operational Activities

Benefits of ERP:-
Integrated Info (Housed 1 location)
Customer Service (Customer info + history)
Security of data.
Flexibility (Work from anywhere)
Centralised data (Promote inter departmental collaboration).
Productivity (Eliminate redundant processes).

LUCKY WRITE SERIES

Date

MIS (Management Info Sys) → Helps organisation "do right things"

DSS (Decision Support System) → Make effective decisions
Helps manager

Semi-Structured :-
Not all info has identical structure.
Some attributes may exist in some entities not in others.

Database System Consists of
1) Data
2) Software
3) Hardware
4) Users

CSMA-CD (sense) :- → collision detection.
↳ Multiple Access.
↳ Carrier (Any shared transmission medium). Carries data.

Date

OS:-

System Software (Allows running other Apps)
Intermediary b/w User and computer.

- Resource Mangement
- Memory "
- Provide Interface. → convenient.

One or More CPU's
Devices connected through bus.
↓
Shared memory

Memory Controller:-

Synchronise Access to memory.

Structure
- Registers
- Cache
- Main Memory. } Volatile
- SSD
- HDD
- Optical disk
- Magnetic tapes.

(Lowest speed + cost)

Types:-

- Batch
- Multitasking
- Multiprogramming
- Multiprocessing.
- Distributed OS

Don't interact with Comp directly.
Each user prepares job offline (punch cards)
All jobs submitted
comp do one by one.

Using many CPU's
Allocated tasks b/w them.

Multiprogramming (old)

(Multiple programs)
One program as a whole keeps running till blocked.

Multitasking:- (New)

time sharing best manifested since each process takes its own share.

Executes multiple by switching in b/w them.

LUCKY WRITE SERIES

UI (GUI, CLI)

Source Code
Compiler
object code
Execute
Output

Date

OS does:-
- UI
- Program Execute
- I/O operation
- File system manipulation
- Resource Allocation.

BIOS:- firmware. (when computer first turned on)
Identify, test and initialise system devices.

Sets machine in state that can give control to OS.

Bootloader:-
- Called by BIOS
- Initialises OS
- Stored in ROM.

locates and load OS kernel to memory

HTML :-

Date

<!DOCTYPE html>
<html>
<head>
<Title> </Title>
<Style>
css
</head> </Style>

header
body
footer
end

<h1>
<p>

<hr> → Horizontal line

 → line break

Tables:-

<table> ↗ tag

<tr> row
<th> header
<td> data

rowspan = "4"
colspan = "3"

Images:-

<img src=" " alt=" ">

Hyperlinks:-

Stylesheet:-

<link rel="Stylesheet" href=" ">

text-align: center;
color:
background-color:

LUCKY WRITE SERIES

Hyper text Markup language.

Date

Not case Sensitive

Relative URL :-

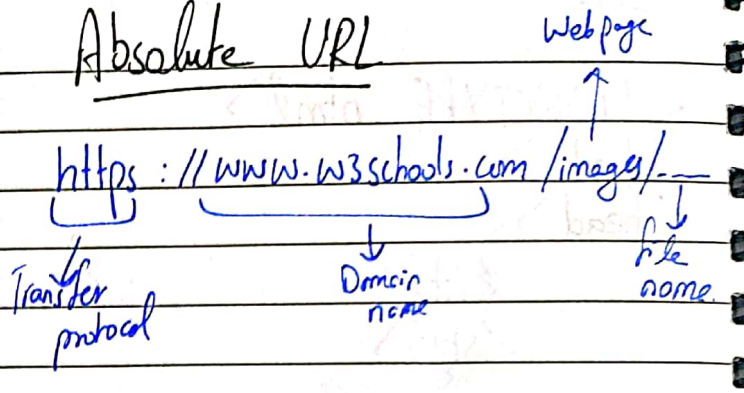
src = "apples.png"

style = "color: red;"

style = "font-family: verdana;"

style = "border: 2px solid red;"

Absolute URL



<i> → italic

<del>

<ins> → inserted text

<sub>

<sup>

red → blue

Color assigning

rgb( )

hsla( )

#

To use an image as a link

Put image tag inside a tag.

<!-- ---- --> Comment

Lists:- Small bullets:- → unordered →

 coffee

ordered numbered

 coffee

Date

In the start use

<Style>
.city {
}
</Style>

→ class

#Ali {
}

→ id

Class vs id:-

→ can only be used by 1 html element

Multiple html elements.

Forms:-

<form>
<label for=" "> → checkbox, submit button.
<input type = "text" id="">
</form>

LUCKY WRITE SERIES